Technical Report: Public Metocean Data & Advisory Consumption Forecasting

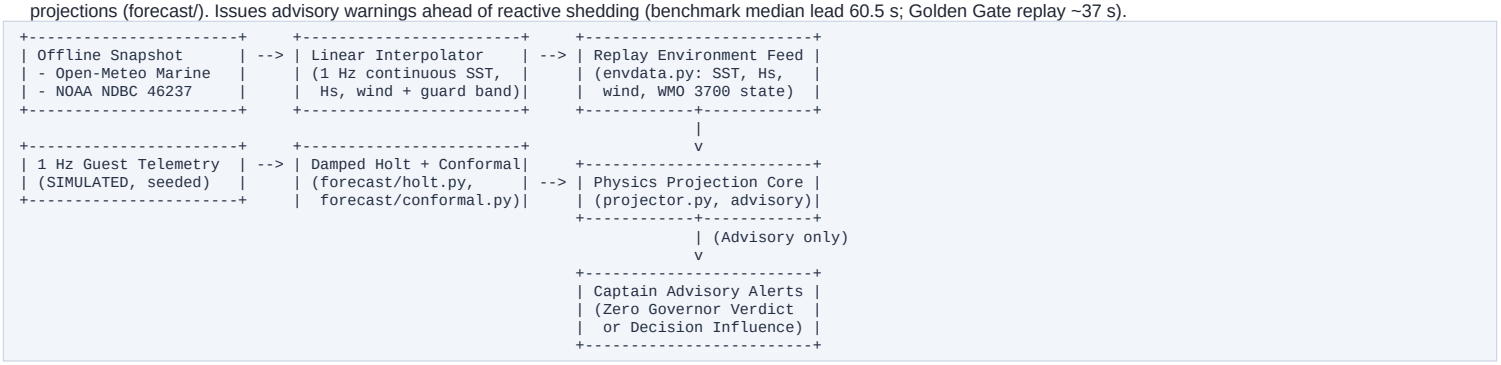
This technical note documents the mathematical formulation, operational architecture, verification gates, and honest operational boundaries of the public environmental data ingestion and advisory load forecasting subsystem (sdv-resource-governor).

1. Architectural Architecture & Non-Interference Boundary

The forecasting and public environmental subsystem consists of two decoupled functional blocks designed for offline, deterministic naval governor execution:

* 1. Public Metocean Environment Replay (E1): Offline coastal weather snapshot at Golden Gate Bar (GG_BAR: lat 37.788, lon -122.634) combining Open-Meteo marine models with NOAA NDBC buoy ground truth (46237). Injects recorded sea surface temperature (SST), wave height (H_s), sea state (WMO Code Table 3700), and sustained wind into replays (envdata.py), and computes guarded, fail-closed operational windows over the 48 h outlook (env_windows.py); missing wind or sea state closes the window.

* 2. Predictive Consumption Advisory Observer (E2): Recursive 1 Hz time-series forecaster with conformal uncertainty intervals and exact physical budget projections (forecast/). Issues advisory warnings ahead of reactive shedding (benchmark median lead 60.5 s; Golden Gate replay ~37 s).



> Mandatory Non-Interference Principle (FC-INT-01..03): The observer is instantiated with observer=None by default and executed strictly outside the governor's authoritative state transitions and decision logic. Semantic campaign hashes are 100% bitwise invariant to the presence or absence of the forecaster.

2. Mathematical Formulation & Predictive Physics

2.1 Normalized Damped Holt-Winters Model (forecast/holt.py)

To make 1 kW auxiliary sensors and 45 kW compute containers directly comparable without scaling distortion, power is normalized by declared mode rating: z_t = y_t / declared_w}(t).

The recursive level (\ell_t) and trend (b_t) update equations with damping factor \phi < 1 are:

* \ell_t = \alpha z_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})

* b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)\phi b_{t-1}

* \hat{z}_{t+h|t} = \ell_t + b_t \phi \frac{1 - \phi^h}{1 - \phi} \quad for } h \in \{10, 20, 30, 45, 60\} s}

* \hat{y}_{t+h|t} = declared_w}(t) \cdot \hat{z}_{t+h|t}

State transitions enforce: (a) initial 20-sample WARMUP (ASM-FCST-WARMUP); (b) trend reset upon mode transition (ASM-FCST-MODE-RESET); (c) gap > 3 s} -> STALE} with 1.5x band expansion (ASM-FCST-STALE); (d) gap > 10 s} -> UNAVAILABLE} (ASM-FCST-UNAVAIL); (e) 3 consecutive residual outliers > 4\sigma -> REGIME_BREAK} (ASM-FCST-REGIME).

2.2 Finite-Sample Split-Conformal Prediction Bands (forecast/conformal.py)

For each horizon h, asymmetric non-conformity residuals r = z_{t+h} - \hat{z}_{t+h|t} are tabulated over the calibration split (calib). Finite-sample corrected asymmetric quantiles q_h^{lo} (5th percentile) and q_h^{hi} (95th percentile) provide nominal 90% confidence bands:

\hat{Y}_{t+h|t} = <=ft[\hat{y}_{t+h|t} + declared_w} \cdot q_h^{lo}, \; \hat{y}_{t+h|t} + declared_w} \cdot q_h^{hi}\right]

2.3 Three-Step Physical Budget Projection (forecast/projector.py)

* 1. Scalar Pre-Filter (Tick cycle): Evaluates upper bound \hat{y}_{hi}(h) > 1.10 x declared_w} (ASM-FCST-OVERRUN-THRESHOLD). If unbreached, execution cost is < 0.1 ms}.

* 2. Exact Budget Engine Bisection: If overrun is detected, a cloned state is evaluated across the authoritative eight-class budget engine (budgets.py). Bisection over h \in (0, 60] s} on the upper 90% trajectory pinpoints the first infeasible horizon h^*_{hi} with at most 8 evaluations; the point/lower crossings are found by a scalar scan against that verified level (no extra engine calls). If the point forecast never reaches it, the output says so (point_forecast_crosses = false).

* 3. Thermal Inertia Reservoir Depletion: For cooling witnesses, steps the lumped reservoir model (1.8 MJ, 30 kW} x 60 s} boost cap) to determine exhaustion second t_{exhaust}}.

3. Pre-Registered Acceptance Gates & Empirical Results

Measured by tools/bench_forecast.py --split both on the production ForecastObserver (on_sample + on_tick), commit c6c1052, macOS arm64. Disjoint seeds: train 1000--1239, calib 2000--2079, test 3000--3119, ood 4000--4119, control_steady_1h 5000--5005. Full output: docs/evidence/forecast_report_20261018.{md,json}. Re-run on the Arduino Ventuno Q (aarch64, same code_sha256 c2408f6f? as Mac, verified by deploy-board): identical metrics, G-LAT p99 2.42 ms (forecast_report_ventunoq.{md,json}; git_commit reads unknown because the board copy has no .git). Thresholds were not changed.

Gate ID	Requirement	Split	Measured	Status
G-NI	Campaign decisions/hash unchanged with observer	pytest non-interference	6 tests green; ebff0e22? unchanged	PASS
G-OFF	Offline runtime, zero network	pytest -k offline	2 tests green	PASS
G-LAT	p99 full observer tick <= 50 ms	test	p99 = 2.42 ms on Ventuno Q (0.83 ms Mac), <= 8 evals/tick	PASS
G-ACC-RAMP	ratio MAE@30 <= 0.85	test	0.785	PASS
G-ACC-STEADY	ratio MAE@30 <= 1.05	test	0.858	PASS
G-COV	90% band coverage in [0.85, 0.95]	test	0.908	PASS
G-RECALL	recall >= 0.90 (WARN active at crossing)	test	1.00	PASS
G-LEAD	median >= 20 s, p10 >= 10 s	test	median 60.5 s, p10 37.9 s	PASS
G-FA	<= 1.0 false WARN / h	test + control 1 h	0.173 / h (3 in 17.33 h)	PASS
G-OOD	recall >= 0.75 and FA <= 3 / h	ood	recall 0.95, FA 9.7 / h	FAIL
G-ENV	false-open windows = 0 with guard	7-day backtest, 169 h	0 (guarded) vs 20 (raw)	PASS

G-OOD fails on false alarms under out-of-distribution profiles: the observer still warns (recall 0.95) but is noisy. All five gates required to publish E2 (G-NI, G-OFF, G-LAT, G-RECALL, G-FA) pass; the OOD failure is declared, not hidden.

4. Golden Gate Replay (what the operator console shows)

GET /api/forecast is a cached REPLAY of scenario_B_golden_gate_recorded.yaml with SIMULATED telemetry (labelled in the payload and UI), not a live feed. Actual alerts:

> "G5 Emergency Comms & Edge Compute Node rising +8 W/s; upper 90% bound reaches 49.5 kW (110% of declared 45.0 kW) in ~39 s; point forecast stays below within 60 s (90% band 39->60 s); binding: power G5 feeder 49.5/49.5 kW. Advisory only -- governor action unchanged." (T = 260 s)

The reactive GUEST_OVERCONSUMPTION fires at T = 297 s exactly as without the observer, so the real advance notice is ~37 s, driven by the upper band (the point forecast alone would not have warned). A second WARN at T = 1540 s with a flat slope is a likely false alarm and is reported as such.

Guarded windows from the base state show no G4 window in the 48 h outlook: VEHICLE_REPLACEMENT_AT_SEA is closed by weather (guarded sea state 4 > limit 3 every hour); LAUNCH_RECOVERY is closed in all 49 hours by operational blockers (G3 acoustic survey quiet, heading not into wind), plus wind > 25 kn in 18 hours. The API/UI print these reasons instead of a fabricated window.

5. Explicit Limitations & Mandatory Honest Disclosures

* 1. Recorded Public Data Snapshot & Offline Runtime: Recorded public data snapshot (UTC 2026-10); runtime is offline with zero live network calls.
* 2. Coastal Model Resolution & Empirical Guard Bands: Open-Meteo is a coastal model product at ~8 km resolution; validated against NDBC 46237/46026/FTPC1 over 7 days (169 hourly pairs); guard bands are the derived 90% error quantiles: +7.34 kn wind, +0.24 m H_s, +0.51 deg C SST (guard_bands.json, status DERIVED). Without the guard the backtest gives 20 false-open hours; with it, 0.
* 3. Synthetic Telemetry & Advisory Non-Interference: Consumption telemetry is synthetic (seeded); /api/forecast is a labelled replay. The forecaster is advisory and never changes the governor’s decision.
* 4. Thermal Baseline Adherence: At Golden Gate SST (~16 deg C), cooling is not the binding constraint; 14-day admission is still judged at 30 deg C design worst-case.
* 5. Known weakness: G-OOD fails (9.7 false WARN / h on out-of-distribution profiles); G-LAT on the Ventuno Q is 2.42 ms p99, far below 50 ms.